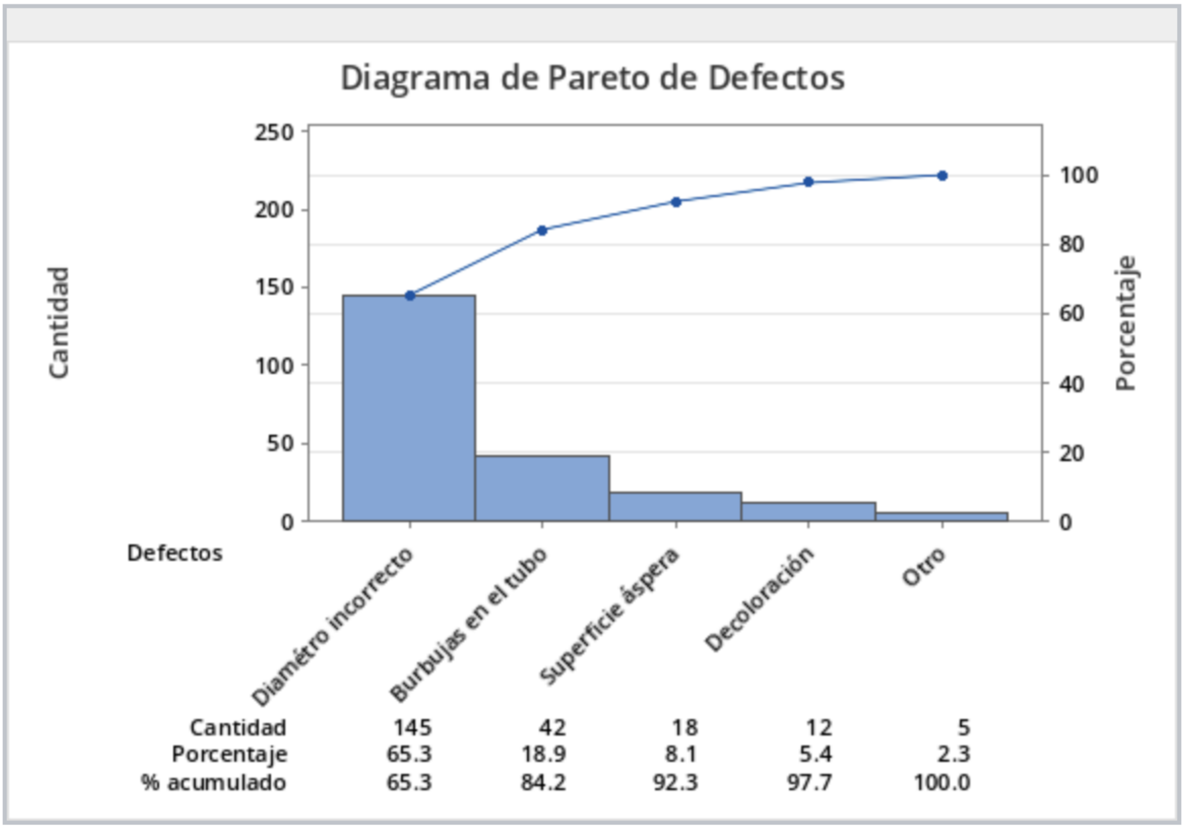
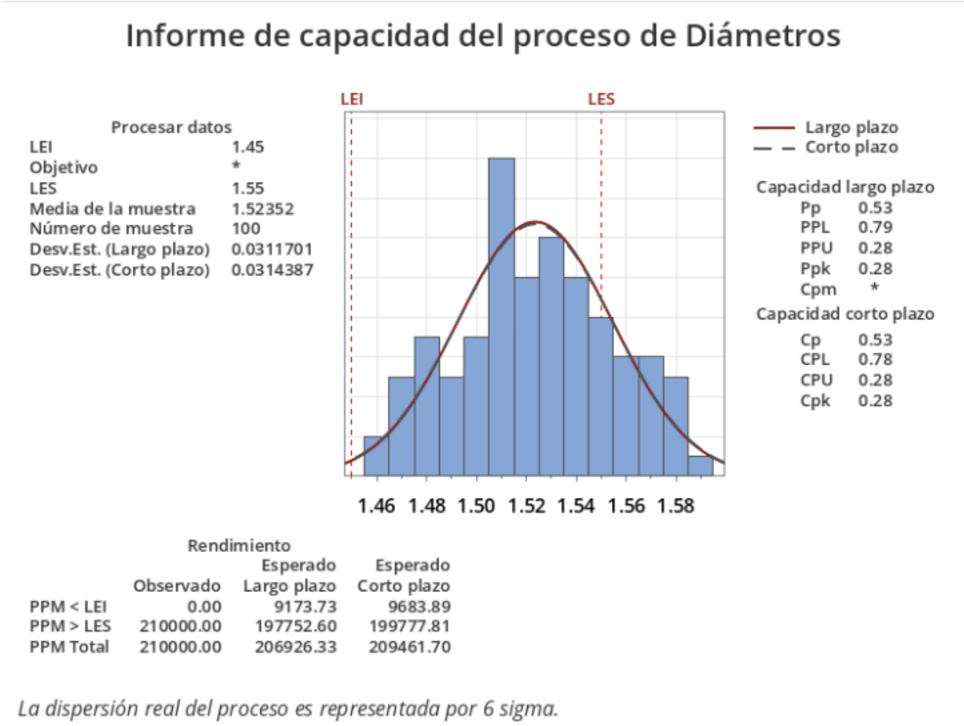
Phase 1: Define & Measure

Objective: Identify the primary source of defects and establish the current baseline capability of the manufacturing process.

Diagrama de Pareto de Defectos



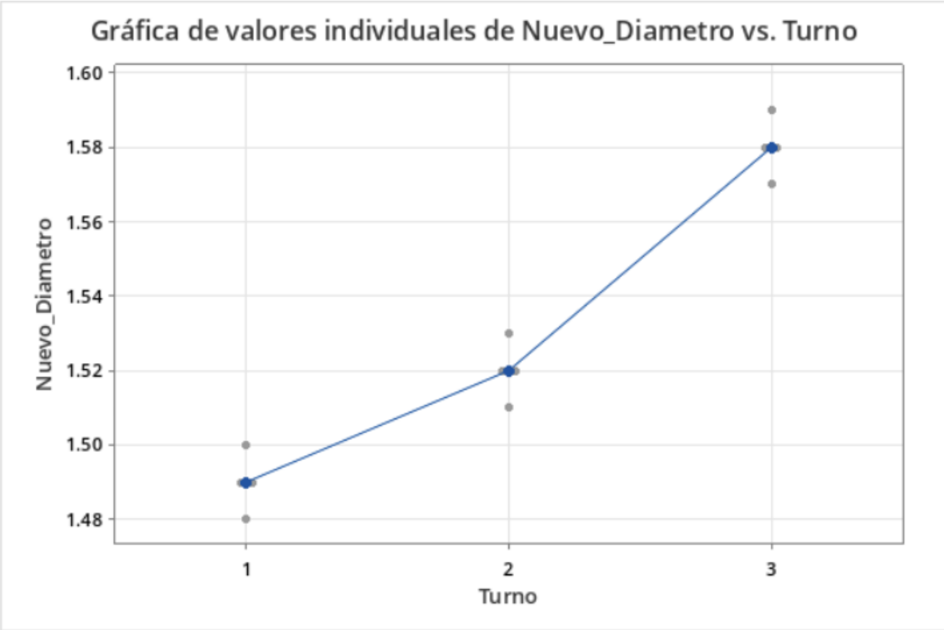
Conclusion: An analysis of historical rejection data indicates that **"Incorrect Diameter"** is the most critical defect, accounting for over 65% of the total rejections. This justifies focusing the optimization efforts on the tube's diameter control.



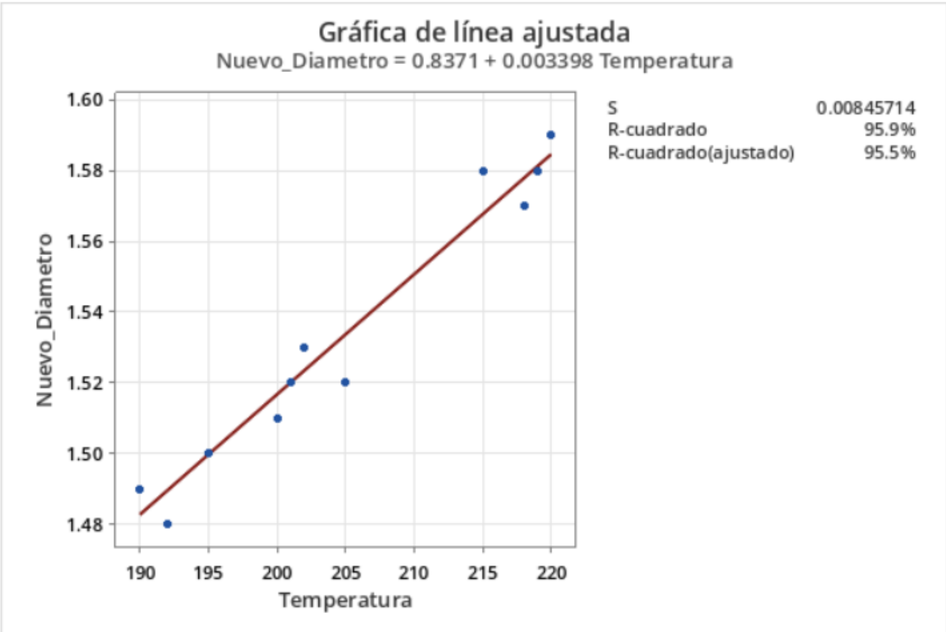
Conclusion: The initial capability study shows that the current process is statistically incapable of meeting the medical specification of 1.50 mm (± 0.05 mm). With a **Cpk < 1.33**, the process data is skewed toward the Upper Specification Limit (USL), confirming that the machine is producing oversized catheters.

Phase 2: Analyze

Objective: Determine the root causes of the diameter variation through statistical hypothesis testing.



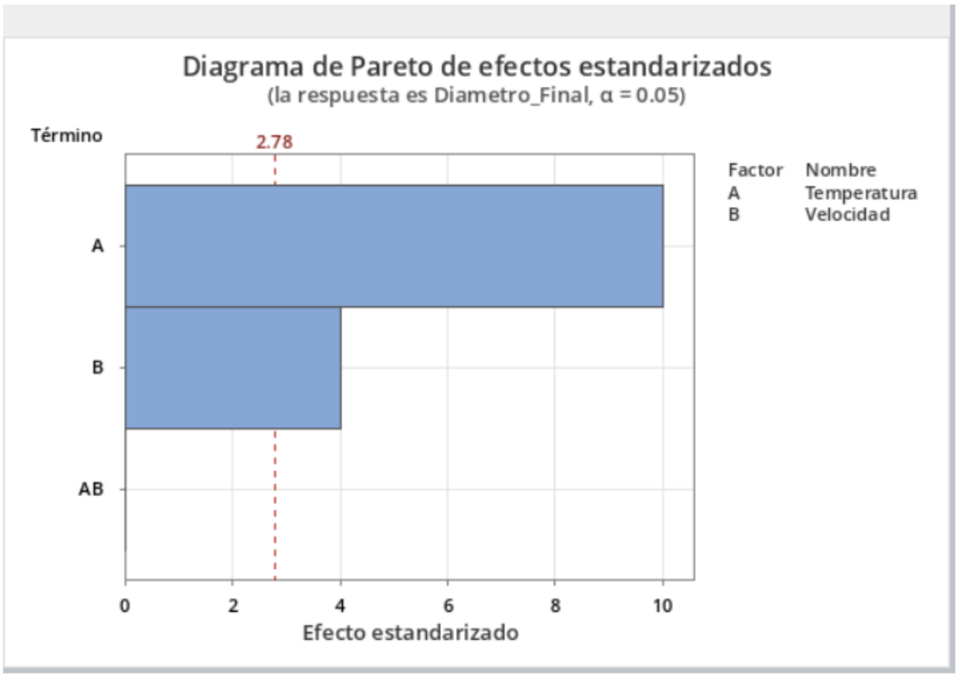
Conclusion: The ANOVA test yields a P-value < 0.05, proving a statistically significant difference between work shifts. The Individual Value Plot clearly demonstrates that **Shift 3** is producing catheters with consistently larger diameters compared to Shifts 1 and 2.



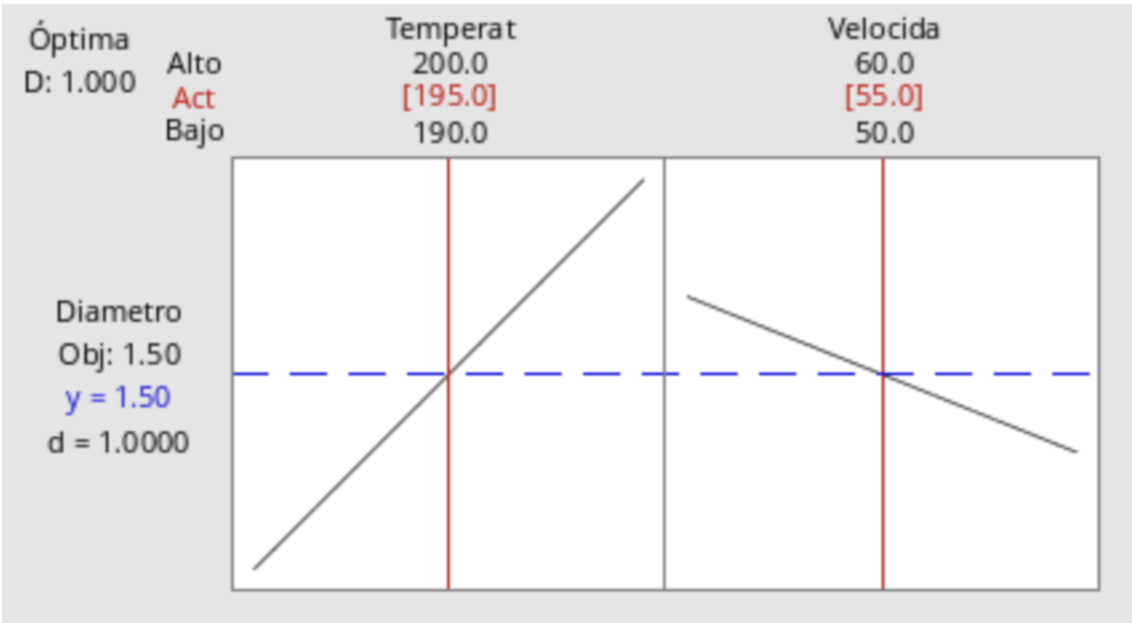
Conclusion: The regression analysis confirms a strong positive correlation between the Extruder Temperature and the catheter's diameter. As temperature increases, the plastic expands further, verifying that temperature control is a critical factor in the defect rate.

Phase 3: Improve

Objective: Execute a Design of Experiments (DOE) to find the exact machine parameters that eliminate the defects.

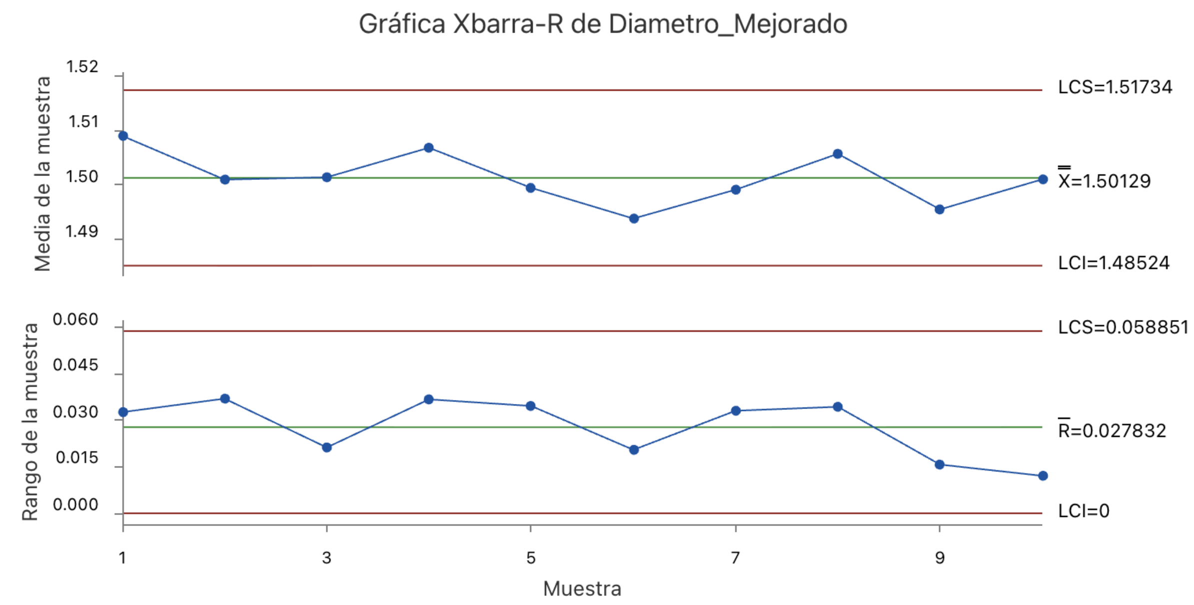


Conclusion: The 2-Level Factorial DOE confirms that both primary factors—**Temperature (Factor A)** and **Extrusion Speed (Factor B)**—have a statistically significant impact on the final diameter, as both bars exceed the critical reference line.



Conclusion: Using the Response Optimizer to target exactly 1.50 mm, the model dictates the optimal operational settings. Adjusting the machine to these specific parameters theoretically minimizes variation and centers the process perfectly on the customer requirement.

Phase 4: Control



Conclusion: Following the implementation of the optimized parameters, a new production run was monitored. The X-bar and R charts show that all data points now fall well within the Upper and Lower Control Limits (UCL/LCL). The process variation has been drastically reduced, and the system is now **stable, centered, and strictly in statistical control.**